Sumangal Bhattacharya, Ph.D.

Date of Birth: May 2, 1994

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Research Interests

- Copula-based modeling of complex dependence structures.
- Multivariate models for early failure and multiple inflated response data.
- Reliability analysis and survival modeling, cure rate models.
- Bayesian inference and advanced computational techniques.
- Nonparametric Inference, U-Statistics, and Jackknife Empirical Likelihood.
- Machine learning methods for statistical modeling and prediction.
- Statistical Learning, Time Series Analysis, and Forecasting.
- Causal Inference and Latent Variable Modeling

Professional Experience

October 2026 – Present — **Assistant Professor, National Institute of Technology Manipur, Imphal, India**.

February 2025 – October 2026 — **Research Associate, Indian Statistical Institute, Delhi Centre, India**.

June 2024 – February 2025 — **Assistant Professor, Shiv Nadar University Chennai, India**.

Academic Record

2019 –2024 — **Ph.D. [*Statistics*]; Indian Institute of Technology Tirupati, India**.
Thesis title: *Copula Based Multivariate Response Models for Correlated Data with Inliers.*
CGPA: *9.0 (Course Work)*

2014 – 2016 — **M.Sc. [*Applied Mathematics*]; Indian Institute of Engineering Science and Technology, Shibpur, India**.
CGPA: *8.55*.

2011 – 2014 — **B.Sc. [*Mathematics (Hons)*]; The University of Burdwan, India.**
Percentage: *63%*.

2010 – 2011 — **Higher Secondary Education** (12th) **[Science]; WBCHSE, India.**
Percentage: *74.2%*.

2008 – 2009 — **Secondary Education** (10th) **; WBBSE, India.**
Percentage: *72.5%*.

National Competitive Examinations

- **CSIR-UGC JRF-2019 (June):** Discipline: *Mathematical Science*; Score: *57.25%*; AIR: *92*.
- **GATE-2016:** Discipline: *Mathematics (MA)*; Score: *477*; AIR: *183*.

Skills

Programming Languages — R, RStan, Python, C++, MATLAB, LaTeX.

Languages — Proficient in reading, writing, and speaking English, Bengali, and Hindi.

Research Articles

- **Multiple Inflated Negative Binomial Regression for Correlated Multivariate Count Data**.
  J. Mathews, **S. Bhattacharya**, S. Sen, I. Das. *Dependence Modeling*, 10(1), 290–307, 2022.
  DOI: 10.1515/demo-2022-0149.
- **Univariate and Bivariate Inverted Exponential–Teissier Distribution in Bayesian and Non-Bayesian Framework to Model Stochastic Dynamic Variation of Climate Data**.
  D. Thakur, **S. Bhattacharya**, I. Das. *Theoretical and Applied Climatology*, 150(3), 1707–1729, 2022.
  DOI: 10.1007/s00704-022-04238-7.
- **On Modeling Bivariate Lifetime Data in the Presence of Inliers**.
  S. Bhattacharya, I. Das, M. Kunnummal. *Annals of Data Science*, 12(1), 1–22, 2024.
  DOI: 10.1007/s40745-023-00511-2.
- **BMW: Inlier Prone Bayesian Models for Correlated Bivariate Data**.
  S. Bhattacharya, I. Das. *Submitted*, 2025.
- **Testing Equality of Lower Partial Moments Using U-Statistics with Applications to Two-Sample and Paired Data**.
  S. Bhattacharya, I. Dewan, S. K. Kattumannil. *Submitted*, 2026.
- **JEL-Based Estimation of Stress–Strength Reliability with Doubly Truncated Random Variables**.
  S. Bhattacharya, I. Dewan, S. K. Kattumannil. *Work in Progress*, 2026.
- **A Median-Based Aging Intensity Function for Reliability Analysis**.
  S. Bhattacharya, A. K. Nanda, I. Das. *Work in Progress*, 2026.
- **A Spatio–Temporal Latent Variable Model for Distinguishing True Incidence and Reporting Effects in Sexual Violence Data**.
  S. Bhattacharya, N. Joshi. *Work in Progress*, 2026.

References

- **Dr. Ishapathik Das (Ph.D. Supervisor)**
  Associate Professor
  Department of Mathematics and Statistics; Indian Institute of Technology Tirupati
  ishapathik@iittp.ac.in; +91 6281022263.
- **Dr. Isha Dewan (Postdoctoral Supervisor)**
  Professor
  Theoretical Statistics and Mathematics Unit; Indian Statistical Institute, Delhi Centre
  isha@isid.ac.in; +91 9810027300.
- **Dr. N. Balakrishna (Internal Thesis Examiner)**
  Professor
  Department of Mathematics and Statistics; Indian Institute of Technology Tirupati
  bala@iittp.ac.in; +91 9446605682.
- **Dr. Sudheesh Kattumannil (Research Collaborator)**
  Professor
  Statistical Science Unit; Indian Statistical Institute, Chennai
  skkattu@isichennai.res.in; +91 9498021164.
- **Dr. Muralidharan Kunnummal (Research Collaborator)**
  Professor
  Department of Statistics; The Maharaja Sayajirao University of Baroda, Gujarat
  Lmv_murali@yahoo.com; +91 9879596190.
- **Dr. Neeraj Joshi (Research Collaborator)**
  Assistant Professor
  Department of Mathematics; Indian Institute of Technology Delhi
  njoshi@maths.iitd.ac.in; +91 9267931092.

Teaching Experience

Assistant Professor

- **Shiv Nadar University, Chennai (2024–2025)**
  - **Calculus and Numerical Methods (4 credits)** – UG Level
  - **Probability and Statistics (4 credits)** – UG Level
  - **Statistical Inference (4 credits)** – UG Level
  - **Discrete Mathematics and Graph Theory (4 credits)** – UG Level

Teaching Assistant

- **Indian Statistical Institute, Delhi Centre (2025-Present)**
  - **Probability II** – UG Level
    Instructor: Dr. Arnab Chakraborty
  - **Statistics II: Introduction to Inference** – UG Level
    Instructor: Dr. Minerva Mukhopadhyay
  - **Statistics III – Multivariate Data and Regression** – UG Level
    Instructor: Dr. Rituparna Sen
  - **Optimization and Numerical Analysis** – UG Level
    Instructor: Dr. Kaushik Jana

- **Indian Institute of Technology, Tirupati (2019-2024)**
  - **Engineering Mathematics I & II** – UG Level
    Instructors: Dr. Shilpak Banerjee, Dr. Ravinder B.
  - **Probability and Statistics** – UG Level
    Instructor: Dr. Ishapathik Das
  - **Calculus** – UG Level
    Instructors: Dr. Rajesh S., Dr. Shilpak Banerjee
  - **Statistical Inference** – UG/PG Level
    Instructor: Dr. Ishapathik Das
  - **Differential Equations and Matrices** – UG Level
    Instructors: Dr. Durga P. Challa, Dr. Ravinder B.
  - **Stochastic Processes and Time Series** – UG/PG Level
    Instructor: Dr. G. S. R. Murthy
  - **Data Science Programming Lab** – UG/PG Level
    Instructor: Dr. Panchatcharam Mariappan

- **NPTEL Online Platform (2024)**
  - **Probability Theory for Data Science (12 Weeks)**
    Instructor: Dr. Ishapathik Das

Workshops, Conferences, and Schools Attended

- **Annual Foundation School - 1 (AFS-1):**
  Place: Hyderabad Central University, India; Duration: 06.05.2019 – 02.06.2019.
- **Instructional School for Teachers on Advanced Topics in Statistical Learning (IST):**
  Place: Indian Statistical Institute, Kolkata; Duration: 11.07.2022 – 23.07.2022.
- **Indian Symposium on Machine Learning (IndoML-2022):**
  Place: Indian Institute of Technology Gandhinagar, India; Duration: 15.12.2022 – 17.12.2022.
- **International Indian Statistical Association (IISA-2022):**
  Place: Indian Institute of Science, Bangalore, India; Duration: 26.12.2022 – 30.12.2022.
- **Workshop in Statistics and Probability:**
  Place: Indian Institute of Technology Bombay, India; Duration: 03.01.2023 – 06.01.2023.
- **Third International Conference on Applications of Mathematics to Nonlinear Sciences (AMNS-2023):**
  Place: Pokhara, Nepal; Duration: 25.05.2023 – 28.05.2023.
- **Stochastic Geometry Days 2023:**
  Place: The University of Bourgogne (Dijon), France; Duration: 12.06.2023 – 16.06.2023.
- **IST-Probability and Statistics (2023):**
  Place: NISER Bhubaneswar, India; Duration: 10.07.2023 – 21.07.2023.
- **International Workshop on Reliability Theory and Survival Analysis (IWRTSA-2023):**
  Place: MIT Manipal, India; Duration: 20.12.2023 – 22.12.2023.
- **International Workshop on Stochastic Models in Data Science (ISPS-2024):**
  Place: Indian Institute of Technology Tirupati, India; Duration: 20.02.2024.
- **Advances in Data Analytics and Official Statistics:**
  Place: Savitribai Phule Pune University, India; Duration: 24.05.2024 – 26.05.2024.
- **International Workshop on Reliability Theory and Survival Analysis (IWRTSA-2024):**
  Place: Banaras Hindu University, India; Duration: 21.12.2024 – 23.12.2024.
- **International Indian Statistical Association (IISA-2024):**
  Place: CUSAT, India; Duration: 27.12.2024 – 31.12.2024.
- **AIS - Statistical Learning Theory (2025):**
  Place: IISER Kolkata, India; Duration: 12.05.2025 – 24.05.2025.
- **Objective Bayes Methodology Conference (O'Bayes 2025):**
  Place: Athens, Greece; Duration: 08.06.2025 – 12.06.2025.

Reviewer for Journals

- **Statistics and Probability Letters** [2025-Present]: Number of manuscripts reviewed: 1.

Leadership and Organizational Roles

- **Core Committee Member and Season Organizer:** The Statistics Club; IIT Tirupati; *2022 – 2024*.
- **Class Representative:** Mathematics Department; IIEST Shibpur; *2014 – 2016*.

I hereby declare that all the above-mentioned things are true to the best of my knowledge and belief.



Date: November 4, 2026

Place: New Delhi, India.

[Sumangal Bhattacharya]